

Beyond the static Rabi limit: dark-state CTAP for asymmetry-resilient transfer

1 · SSH model & domain walls

The SSH chain has alternating hoppings v (intracell) and w (intercell). For $v < w$ it is a 1D topological insulator ($\nu = 1$) with exponentially localized zero-energy edge states protected by **chiral symmetry**; bulk gap $\Delta = 2|w - v|$, localization length $\xi = -1/\ln(v/w)$.

A **domain wall** hosts three protected states near $E = 0$ — $|L\rangle, |S\rangle, |R\rangle$ — an effective three-site chain:

$$H_{\text{eff}} = J_{LS}|S\rangle\langle L| + J_{SR}|R\rangle\langle S| + \text{H.c.}$$

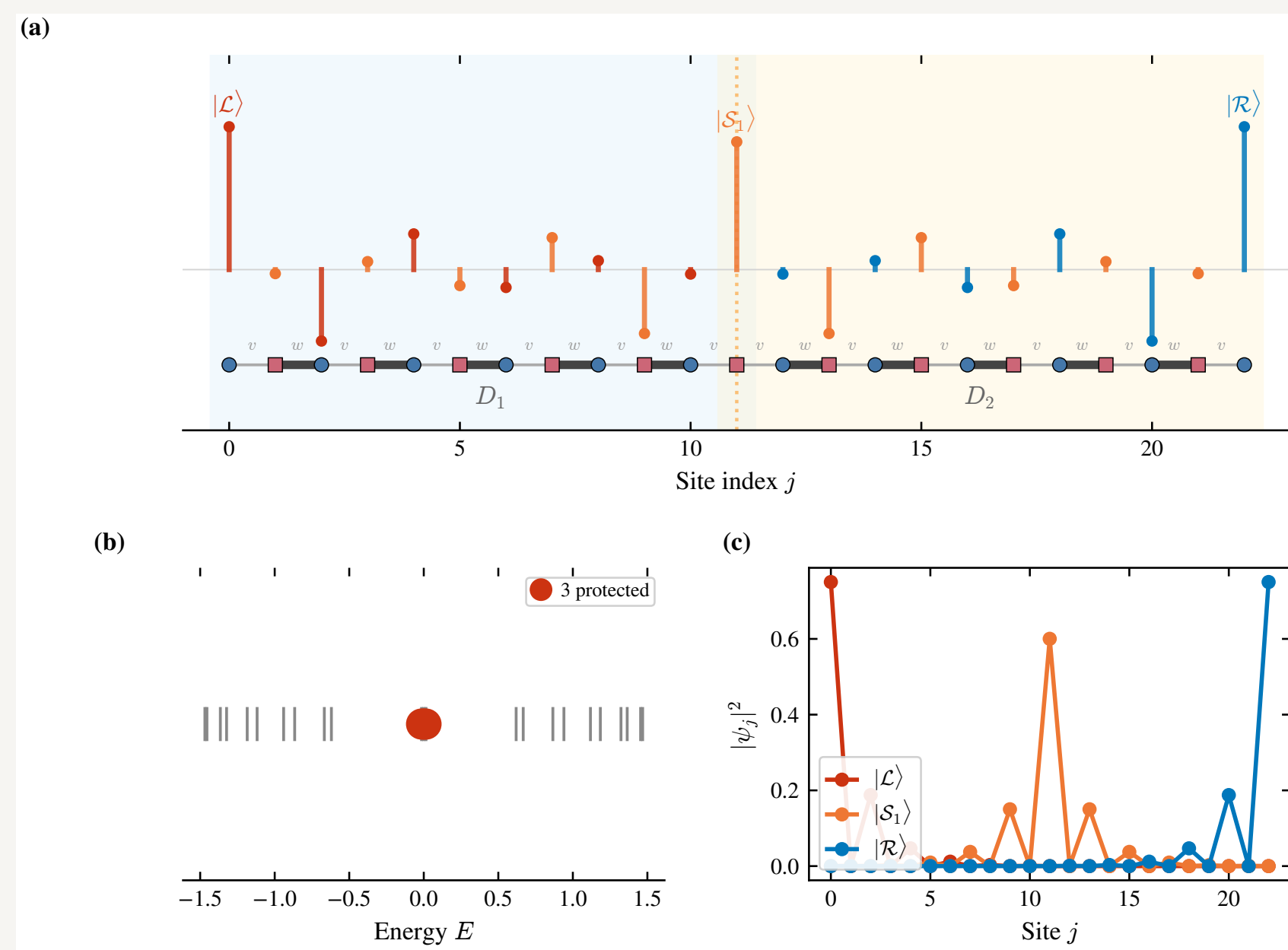


Fig. 1 — three protected states $|L\rangle, |S\rangle, |R\rangle$ in a two-domain SSH chain.

2 · Symmetric transfer works

An adiabatic ramp of $v(t)$ drives $|L\rangle \rightarrow |S\rangle \rightarrow |R\rangle$; fidelity $f = |\langle L-1|\psi(t_{\text{tr}})\rangle|^2$. The transfer time scales exponentially with the domain length,

$$t_{\text{tr}} \sim \frac{\pi}{\sqrt{2}J} \sim \left(\frac{w}{v}\right)^{\ell/2}.$$

Symmetric chain ($L = 11$, $\ell = 4$): $f = 0.999$ at $t_{\text{tr}} = 45$ (analytic 45.6).

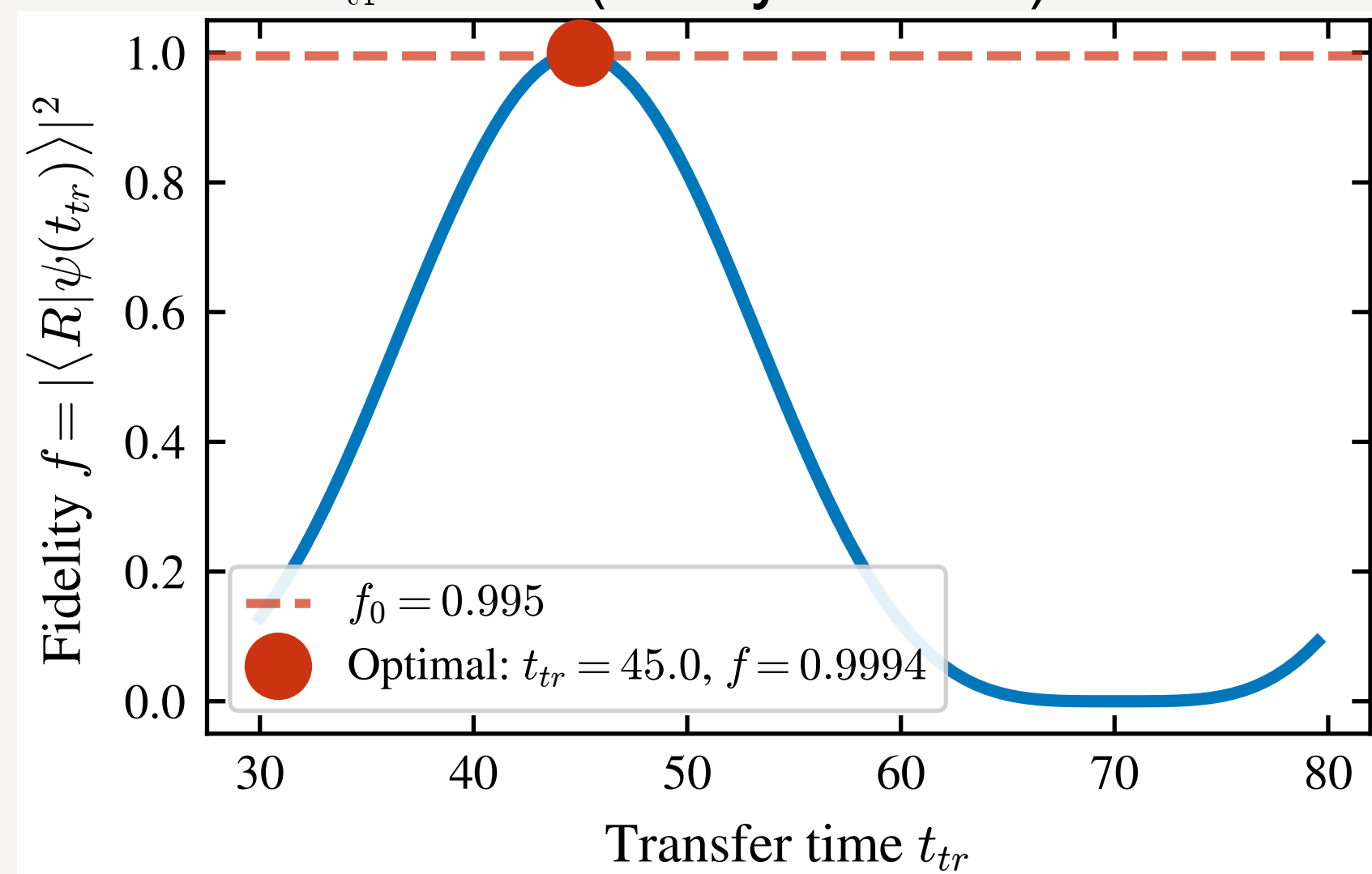


Fig. 2 — transfer fidelity vs t_{tr} ; first peak $f = 0.999$.

3 · Off-centre walls break Rabi

Couplings scale exponentially, $J_{LS}/J_{SR} \sim (w/v)^{(\ell_2-\ell_1)/2}$, and a three-site chain is capped by

$$f_{\text{max}} = \frac{4J_1^2 J_2^2}{(J_1^2 + J_2^2)^2}.$$

Fidelity collapses off-centre:
 $0.637 \rightarrow 0.061 \rightarrow 0.004$.

Shortcut-to-adiabaticity pulses do *not* help — the bottleneck is the coupling ratio in the transfer phase, fixed by geometry.

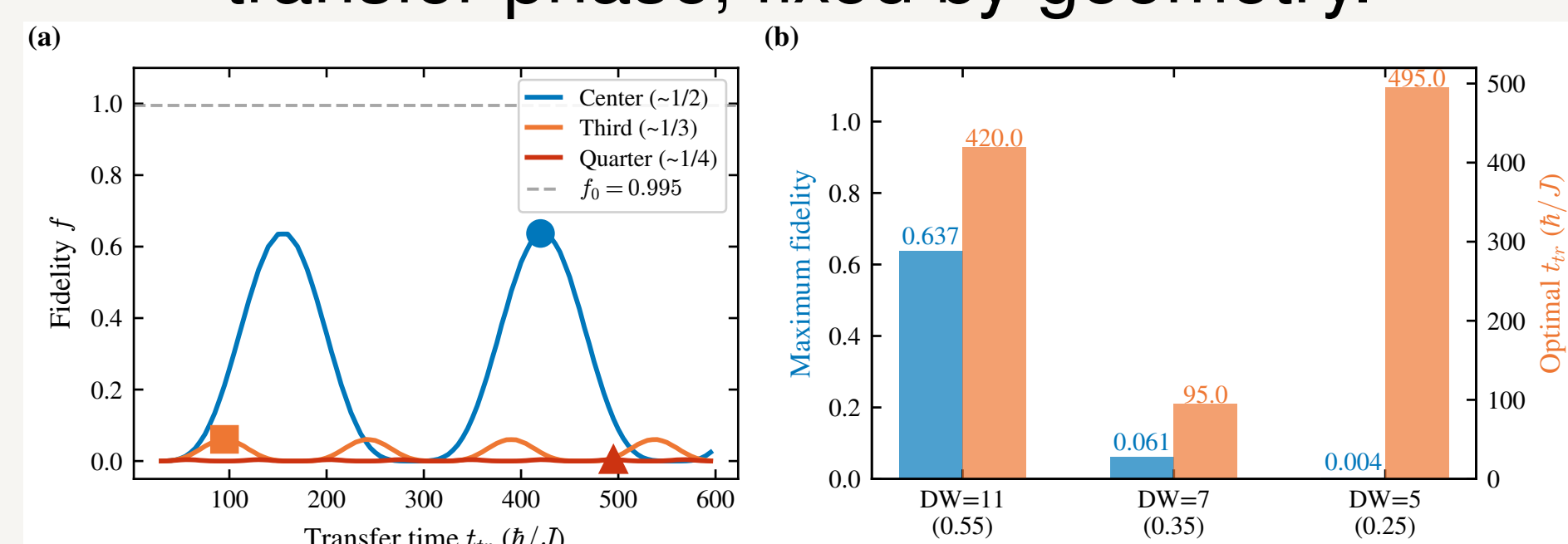


Fig. 3 — peak transfer fidelity vs wall position; collapses off-centre ($0.637 \rightarrow 0.061 \rightarrow 0.004$).

4 · It's the protocol, not topology

The collapse is **not** a failure of topological protection: the edge states stay gapped at every wall position. Give the left and right domains **independent** controls $v_L(t), v_R(t)$. The effective Hamiltonian always has a zero-energy **dark state** with no soliton component:

$$|D_0(t)\rangle = \cos \Theta(t) |L\rangle - \sin \Theta(t) |R\rangle, \quad \tan \Theta = J_{LS}/J_{SR}.$$

Adiabaticity, $\sqrt{J_{LS}^2 + J_{SR}^2} \gg |\dot{\Theta}|$, is met by long enough pulses — *independently of the asymmetry*.

5 · CTAP dark-state passage

Sweeping $\Theta : 0 \rightarrow \pi/2$ transfers $|L\rangle \rightarrow -|R\rangle$ without ever populating $|S\rangle$. Counter-intuitive ordering: J_{SR} fires *before* J_{LS} . A counterdiabatic term $H_{\text{CD}}^{\text{eff}} \propto \dot{\Theta}(|L\rangle\langle R| - |R\rangle\langle L|)$ reaches the same fidelity in arbitrarily short time.

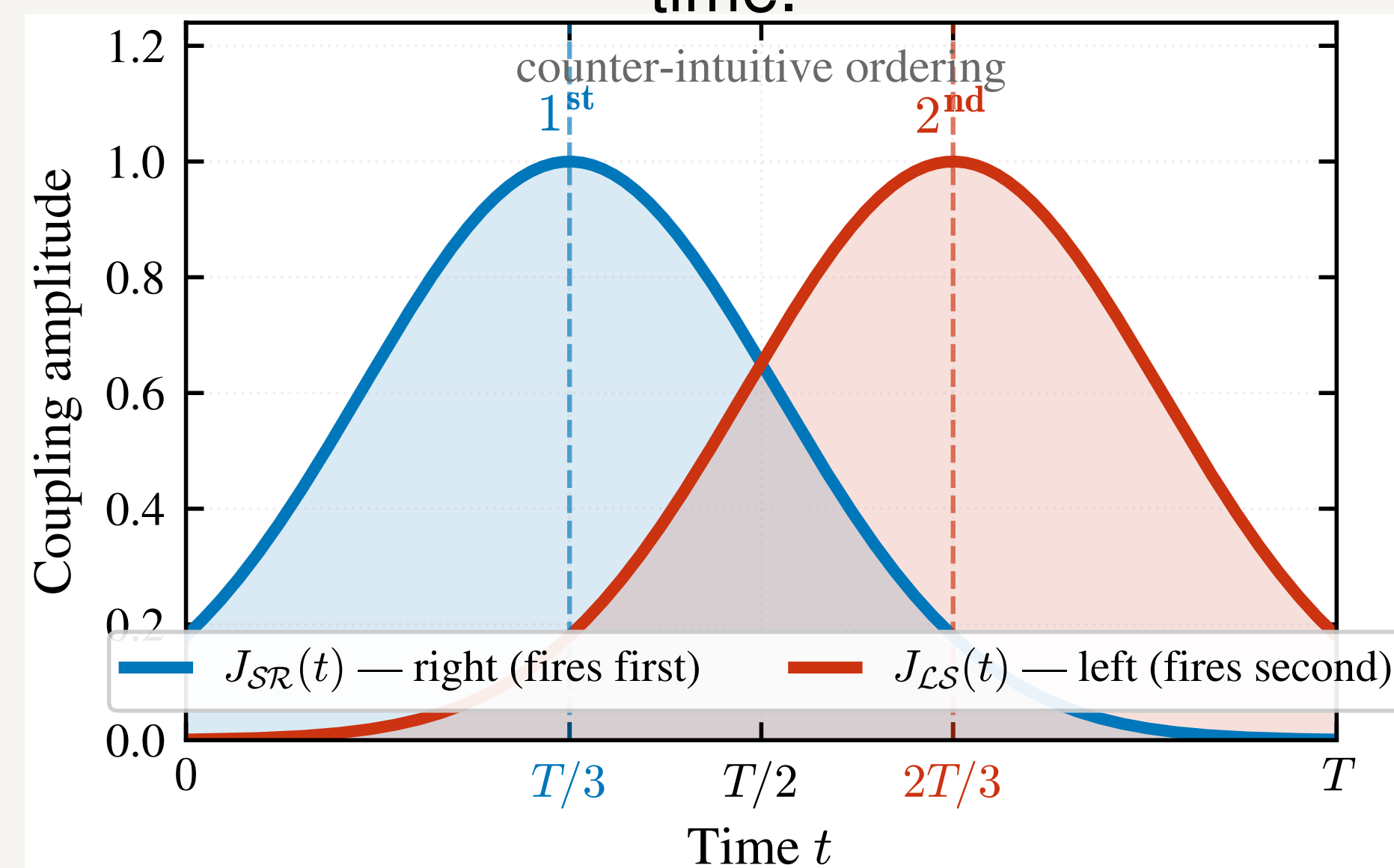


Fig. 4 — counter-intuitive ordering: J_{SR} (Stokes) before J_{LS} (pump).

6 · Full-chain validation ($L = 21$)

j_{DW}	Rabi	CTAP	leakage
11	0.637	0.9998	$< 10^{-7}$
7	0.061	0.9999	$< 10^{-7}$
5	0.004	0.9995	$< 10^{-7}$

The 3×3 reduction is exact to $< 0.025\%$. The only residual infidelity is a removable ceiling $f \leq \cos^2 \Theta_0$ — not a topological bound.

Asymmetry is paid in **time**, not fidelity.

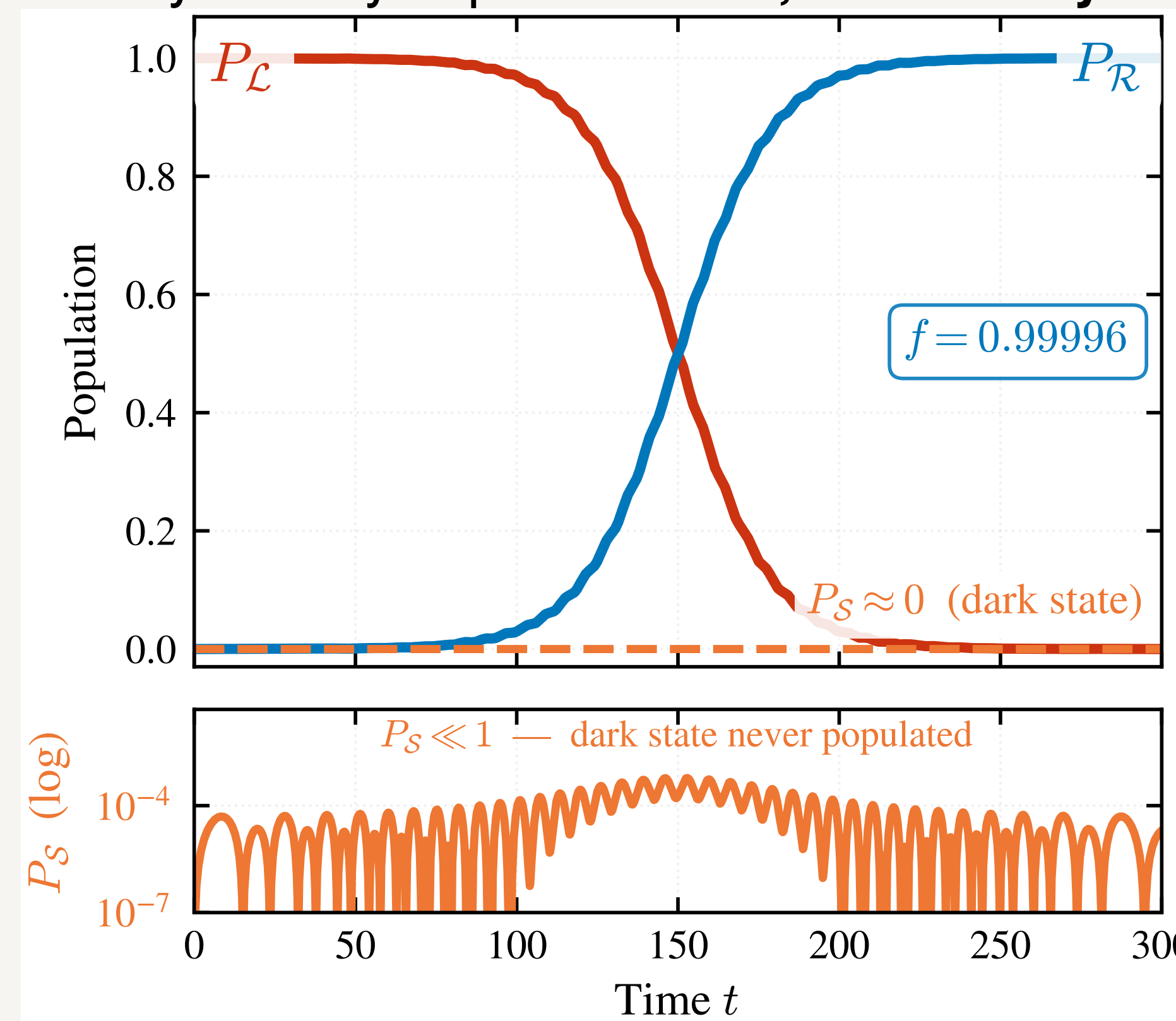


Fig. 5 — dark-state transfer: $P_L \rightarrow P_R$ with $P_S \approx 0$ ($f = 0.99996$).

7 · Collective CTAP speed-up

Single-wall CTAP is slow (T grows exp. with the longest domain). Subdividing into D domains shrinks the bottleneck segment ($\ell_s \simeq L/D$, $J_{\text{bott}} \sim (v/w)^{\ell_s/2}$): up to $\times 13134$ at $D = 4$. A single collective sequence drives the $D = 4$ chain ($L = 45$) with $F = 0.998$, NN couplings only.

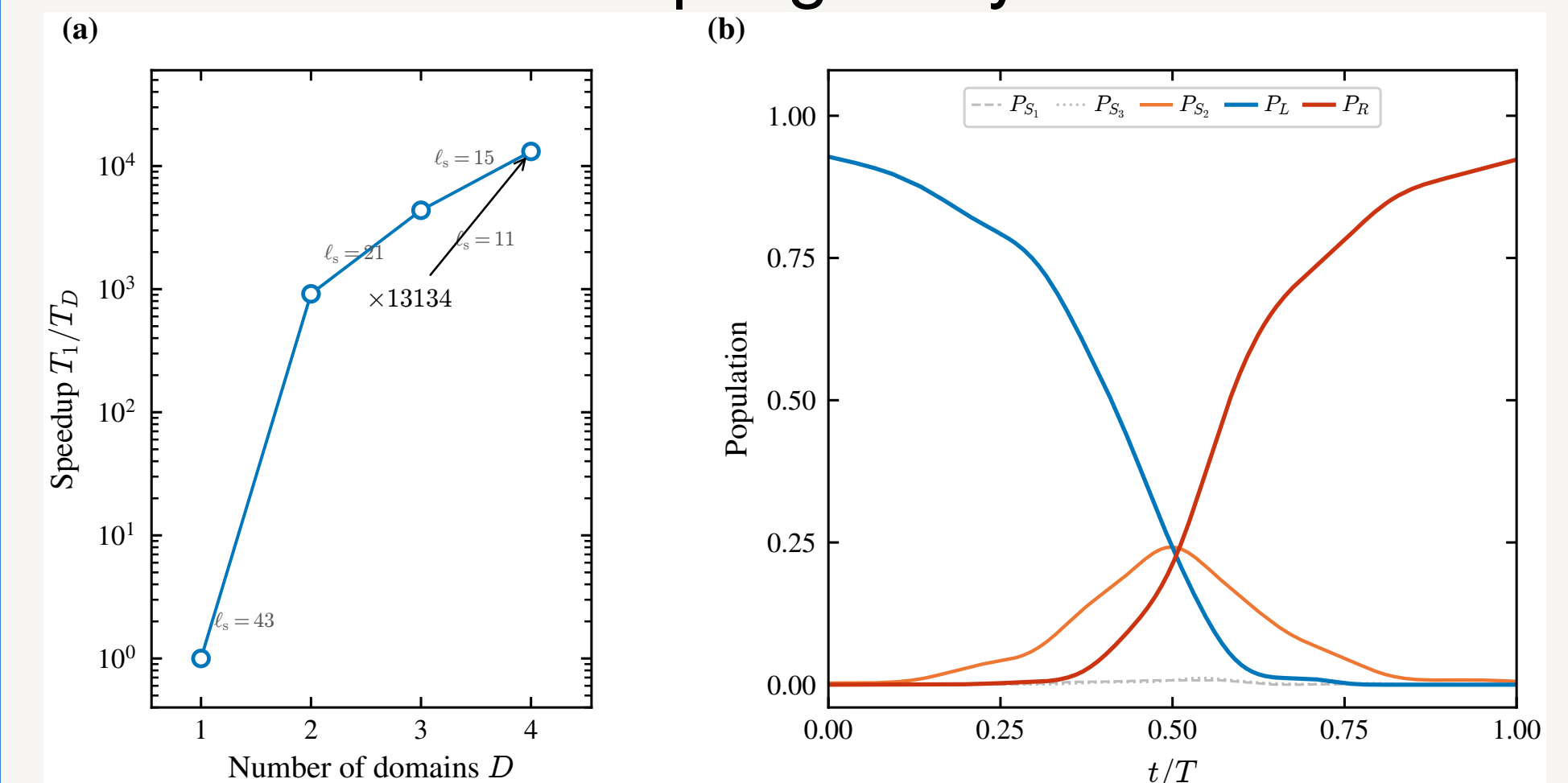


Fig. 6 — speed-up vs D and collective five-state transfer ($F = 0.998$).

8 · Chiral symmetry & robustness

Off-diagonal disorder respects chiral symmetry: the zero mode stays pinned at $E = 0$ to $\sim 10^{-16}$ and CTAP fidelity stays flat at ≈ 0.99 . On-site disorder and NNN hoppings break it ($\langle |E_0| \rangle \approx 0.95 \kappa$); on-site disorder drives $f \rightarrow 0.30$ at $\sigma = 0.20$. CTAP (NN) **inherits** the protection; the CD protocol (NNN) **forfeits** it. CTAP is also $\sim 8\times$ more tolerant to wall misplacement.

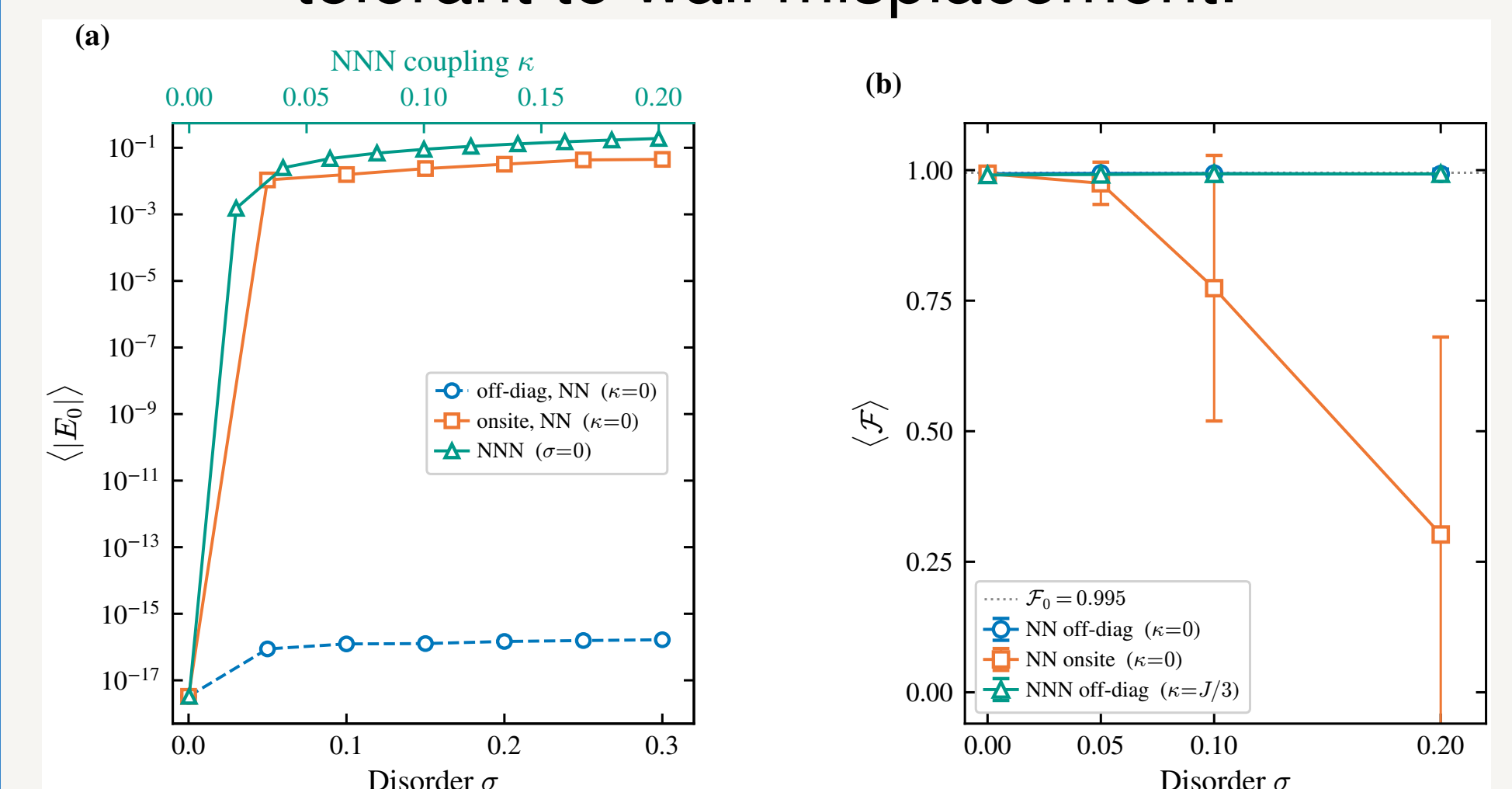


Fig. 7 — zero-mode energy and CTAP fidelity under disorder.

9 · Conclusions

- Domain-wall transfer reproduced: $f = 0.999$ (symmetric).
- Static Rabi fails off-centre — a protocol artifact, not a topological limit.
- CTAP recovers $f > 0.999$ at any wall position; leakage $< 10^{-7}$.
- Collective multidomain CTAP: $F = 0.998$, > 4 orders speed-up.
- **The control protocol — not topology alone — decides robustness.**
- Two-wall transfer needs straddling CTAP (even-parity constraint).

10 · Outlook

- Straddling CTAP (SCTAP) for two-domain-wall chains.
- Hybrid domain-wall + CD driving for the time-critical ramps.
- Platforms: superconducting qubits, cold atoms, quantum dots.

